Project Summary: University Admission Prediction

1. Project Objective

The goal of this project was to develop a machine learning model to predict a student's "Chance of Admit" to a graduate program based on their academic profile. The project involved exploring the data, comparing multiple regression models, and identifying the key factors that influence admission.

2. Methodology

- **Data:** The project used the "Graduate Admission" dataset, which contained 500 student records with 7 predictive features (GRE, TOEFL, University Rating, SOP, LOR, CGPA, Research) and one target variable (Chance of Admit).
- **Preprocessing:** The data was very clean, with **no missing values**. The only preprocessing step was to drop the irrelevant `Serial No.` column.
- **Data Splitting:** The dataset was split into an **80% training set** (400 samples) for model training and a **20% testing set** (100 samples) for final evaluation.

3. Model Comparison

Four different regression models were trained and evaluated to find the best performer. The primary metric for comparison was the **R-squared (R^2)** score on the unseen test set.

Model	R-squared (R^2) Score	Mean Absolute Error (MAE)
Linear Regression	**0.8188**	**0.0427**
Ridge Regression	0.8180	0.0429
Lasso Regression	0.8138	0.0435
Random Forest	0.7875	0.0454

Finding: The simple **Linear Regression** model achieved the highest R^2 score, indicating it was the most accurate and reliable model for this dataset.

4. Hyperparameter Tuning

To ensure a thorough evaluation, `GridSearchCV` (with 5-fold cross-validation) was performed on the `Ridge Regression` model. This tuning process confirmed that even the best-tuned Ridge model (R^2: 0.8180) could not outperform the baseline Linear Regression model.

5. Final Model & Key Insights

- **Final Model:** `LinearRegression`
- **R-squared: 0.8188** (The model can explain ~81.9% of the variance in admission chances.)

- **Mean Absolute Error (MAE): 0.0427** (On average, the model's prediction is only off by ~4.3%.)

A feature importance analysis was conducted to understand what the model learned.

Key Insights:

- **CGPA** is by far the **most important factor** in predicting admission.
- GRE Score and TOEFL Score are the second and third most important factors.
- Research, LOR, SOP, and University Rating have a positive but less significant impact.

6. Conclusion

This project successfully built and evaluated several models to predict university admissions. The **Linear Regression** model was selected as the best performer due to its high accuracy. The key takeaway is that an applicant's academic performance, especially their **CGPA**, is the dominant factor in predicting their admission chances.